

# RCIM Model-Bank Reproduction Remaining Family Residual Cellwise Closure Campaign Results Report

## Feedforward Transmission Error Baseline

StyLED PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

## Overview

This report closes the residual exact-paper RCIM Model-Bank Reproduction retry wave prepared in:

- `doc/reports/campaign_plans/track_1/exact_paper/2026-04-19-01-04-28_track1_remaining_family_residual_cellwise_closure_campaigns_plan_report.md`  
The batch targeted every non- SVM family-harmonic pair that still remained `δYi` or `δYj` after the earlier 171 -run cellwise reference wave.
- completed family campaigns: 9
- completed validation runs: 1026
- failed validation runs: 0
- promoted residual pair winners: 90/171
- aggregate campaign artifact root:  
`output/training_campaigns/track1/exact_paper/forward/residual_cellwise_closure/shared/track1_remaining_family_residual_cellwise_closure_campaigns_2026_04_19_01_04_28/`

## Operational Outcome

- every residual family campaign now exposes:
  - `campaign_leaderboard.yaml`
  - `campaign_best_run.yaml`
  - `campaign_best_run.md`
- the aggregate residual campaign root now exposes the final 1026 -entry leaderboard and deterministic bookkeeping representative
- `doc/running/active_training_campaign.yaml` now records the canonical results report path for the completed residual retry wave
- the canonical benchmark now reads from the best accepted pair winner across:
  - the earlier 171 -run `cellwise_reference` wave
  - the newer 1026 -run residual retry wave



## Family Recovery Outcome

The table below uses compact display aliases for PDF fit. The exact canonical run names remain serialized in the family campaign leaderboards and best-run artifacts under `output/training_campaigns/`.

| Family | Amplitude Run | Phase Run | Best Run | Best Scope      | Best Closure Score | Open Cells |
|--------|---------------|-----------|----------|-----------------|--------------------|------------|
| MLP    | A162 a01      | P156 a01  | A162 a01 | amplitudes_only | 1.000              | 0          |
| RF     | A0 a03        | P156 a01  | A0 a03   | amplitudes_only | 1.000              | 0          |
| DT     | A0 a02        | P156 a01  | A0 a02   | amplitudes_only | 1.000              | 0          |
| ET     | A0 a01        | P156 a01  | A0 a01   | amplitudes_only | 1.000              | 0          |
| ERT    | A0 a02        | P156 a01  | A0 a02   | amplitudes_only | 1.000              | 0          |
| GBM    | A0 a02        | P156 a01  | A0 a02   | amplitudes_only | 1.000              | 0          |
| HGBM   | A0 a03        | P156 a01  | A0 a03   | amplitudes_only | 1.000              | 0          |
| XGBM   | A0 a02        | P156 a01  | A0 a02   | amplitudes_only | 1.000              | 0          |
| LGBM   | A0 a02        | P156 a01  | A0 a02   | amplitudes_only | 1.000              | 0          |

## Aggregate Ranking

The aggregate ranking below also uses compact run aliases for PDF fit. The exact run names remain preserved in the aggregate leaderboard and family-level campaign artifacts.

| Rank | Run      | Family | Scope           | Paper Cell | Met | Near | Open | Closure Score |
|------|----------|--------|-----------------|------------|-----|------|------|---------------|
| 1    | A0 a02   | DT     | amplitudes_only | 2          | 2   | 0    | 0    | 1.000         |
| 2    | A0 a05   | DT     | amplitudes_only | 2          | 2   | 0    | 0    | 1.000         |
| 3    | A156 a02 | DT     | amplitudes_only | 2          | 2   | 0    | 0    | 1.000         |
| 4    | A162 a02 | DT     | amplitudes_only | 2          | 2   | 0    | 0    | 1.000         |
| 5    | A1 a02   | DT     | amplitudes_only | 2          | 2   | 0    | 0    | 1.000         |
| 6    | A1 a05   | DT     | amplitudes_only | 2          | 2   | 0    | 0    | 1.000         |
| 7    | A240 a01 | DT     | amplitudes_only | 2          | 2   | 0    | 0    | 1.000         |
| 8    | A240 a02 | DT     | amplitudes_only | 2          | 2   | 0    | 0    | 1.000         |
| 9    | A240 a03 | DT     | amplitudes_only | 2          | 2   | 0    | 0    | 1.000         |
| 10   | A240 a04 | DT     | amplitudes_only | 2          | 2   | 0    | 0    | 1.000         |
| 11   | A240 a05 | DT     | amplitudes_only | 2          | 2   | 0    | 0    | 1.000         |
| 12   | A240 a06 | DT     | amplitudes_only | 2          | 2   | 0    | 0    | 1.000         |
| 13   | A39 a01  | DT     | amplitudes_only | 2          | 2   | 0    | 0    | 1.000         |
| 14   | A39 a02  | DT     | amplitudes_only | 2          | 2   | 0    | 0    | 1.000         |
| 15   | A39 a03  | DT     | amplitudes_only | 2          | 2   | 0    | 0    | 1.000         |

## Final Batch Winner

The aggregate batch bookkeeping representative is `track1_dt_amplitude_0_closure_attempt_02` from family `DT`. This is a deterministic winner under the per-run closure-first policy across the full `1026`-entry residual batch. It should be read as a bookkeeping representative of the completed campaign, not as the whole scientific verdict of the promoted pairwise residual upgrades.

- winning scope: `amplitudes_only`
- closure score: `1.000`
- met paper cells: `2`
- near paper cells: `0`
- open paper cells: `0`

## Scientific Outcome

- the residual retry wave materially improved the accepted benchmark surface *without changing the exact-paper family definitions*
- amplitude `RMSE` closure rises to `10/10`
- phase `MAE` closure rises to `9/9`
- phase `RMSE` closure rises to `8/9`
- amplitude `MAE` closure rises to `7/10`
- no harmonic remains fully red in Table `6`
- `RCIM Model-Bank Reproduction` remains canonically open because the harmonic-level family alignment rule is still not fully satisfied

## Final Benchmark Refresh Effect

The canonical benchmark after the residual retry closeout is:

- Table `2` amplitude `MAE` : `7/10` harmonics meet or beat the paper target
- Table `3` amplitude `RMSE` : `10/10` harmonics meet or beat the paper target
- Table `4` phase `MAE` : `9/9` harmonics meet or beat the paper target
- Table `5` phase `RMSE` : `8/9` harmonics meet or beat the paper target
- harmonic-level Table `6` closure: `2/10` fully matched, `8/10` partially matched, `0/10` not yet matched The highest-priority still-open harmonics are now:
  - `0`
  - `3`
  - `39`
  - `78`
  - `81`
  - `156`
  - `162`
  - `240`

## Final Interpretation

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This batch is operationally complete and materially stronger than the earlier non- SVM cellwise reference wave.

The key final conclusion is

- the residual retry wave is fully closed out
- the repository now has the aggressive 1026 -run residual retry evidence on top of the earlier 171 exact-paper reference runs
- the canonical benchmark and Tables 2-5 now read from the best accepted pair winner across both waves
- RCIM Model-Bank Reproduction is no longer numerically blocked by broad red surfaces; the remaining work is concentrated on harmonic-level family alignment in Table 6